



B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19
(Autonomous Institute, Affiliated to VTU)
Department Name: CSE

THIRD INTERNALS

Course Code: 23CS6PCMAL	Course Title: Machine Learning	
Semester: 6 th	Maximum Marks: 40	Date:
Faculty Handling the Course:	Rekha G S, Dr.Kayarvizhy N, Dr. Pallavi G B, Saritha A N	
Instructions: <i>Internal choice is provided in Part C.</i>		

PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO	PO	BL
1a	Explain generalization in Reinforcement Learning with an example • Generalization refers to the ability of an RL agent to perform well in unseen or similar states after learning from limited experiences. • Instead of memorizing exact states, the agent learns patterns. • Helps in large or continuous state spaces. • Example: A robot trained to navigate one room can navigate a similar room.	5M			

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO	PO	BL
2a	An agent is learning the value of states using Temporal Difference (TD(0)) learning. During one step of interaction, the agent observes the following: Current state value: $V(s) = 10$, Next state value: $V(s') = 12$ and Immediate reward: $r = 2$. The learning parameters are: Learning rate $\alpha = 0.5$ and Discount factor $\gamma = 0.9$ (a) Compute the updated value $V(s)$ using the TD(0) update rule. Determine whether the value of the state increases or decreases after the update. (b) Justify your answer by interpreting the role of reward and next state value in the update process. Given: Current state value: $V(s)=10$ Next state value: $V(s')=12$ Reward: $r=2$ Learning rate: $\alpha=0.5$ Discount factor: $\gamma=0.9$ Substitute the values into the formula: $V(s)=10+0.5[2+0.9(12)-10]$ Updated state value: $V(s)=11.4$	5M			

	Since the value changed from 10 to 11.4, the state value increases after the update. (b) Interpretation - The immediate reward $r=2$ provides positive feedback to the agent, encouraging the current state. - The next state value $V(s')=12$ is high, meaning the future state is expected to give good long-term rewards. - The discount factor $\gamma=0.9$ allows future rewards to strongly influence learning. - Therefore, both the positive reward and valuable next state increase the estimated value of the current state. Hence, the agent updates the current state toward a better expected future return.				
2b	In a Q-learning algorithm, the Q-value is updated using the difference between the current estimate and the target value. During an update step, the following values are observed: Old Q-value: $Q_{old} = 4$ and Target value: $Q_{target} = 10$ (a) Compute the updated Q-value for the following learning rates: (i) $\alpha = 0.1$ and (ii) $\alpha = 0.9$. Compare the results obtained in both cases. (b) Analyze the trade-off between learning speed and stability for small and large learning rates. (i) When $\alpha=0.1$ Updated Q value : $Q_{new}=4.6$ When $\alpha=0.9$ $Q_{new}=9.4$ Comparison of Results With $\alpha=0.1$, the Q-value changes slowly from 4 to 4.6. With $\alpha=0.9$, the Q-value changes rapidly from 4 to 9.4, moving much closer to the target value 10. Thus, a larger learning rate causes faster adaptation to new information.	5M			
2c	Two reinforcement learning agents operate in the same environment but use different discount factors: Agent A: $\gamma = 0.2$ and Agent B: $\gamma = 0.9$. Both agents expect a future reward of 100, which will be received after 3-time steps. Compute the discounted reward for both Agent A and Agent B. Compare the results obtained for the two agents. Analyze how the choice of discount factor γ affects the agent's behavior in terms of short-term vs long-term decision making The discounted reward is calculated using: Discounted Reward $= \gamma t \times R$ $\text{\text{Discounted Reward}} = \gamma^t \times R$ Where: $R=100$ (future reward) $t=3t$ time steps γ= discount factor Agent A ($\gamma=0.2$) Discounted Reward $= (0.2)^3 \times 100$	5M			

	$=0.008 \times 100$ $=0.8$ Agent B ($\gamma=0.9$) Discounted Reward $= (0.9)^3 \times 100$ $=0.729 \times 100$ $=72.9$ Agent A gives very little importance to the future reward. Agent B values the future reward much more strongly. Thus, Agent B considers long-term benefits more effectively than Agent A.				
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PART- C

Total 20 Marks (Choice)

No.	Question	Marks	CO	PO	BL																																																
3a	An educational institution wants to analyze student learning patterns using unsupervised learning, specifically the K-Means clustering algorithm, to group students based on engagement rather than direct performance scores. The dataset collected is as follows: <table> <tr> <th>Student</th> <th>Study Hours (per day)</th> <th>Attendance (%)</th> <th>Marks</th> </tr> <tr> <td>S1</td> <td>2</td> <td>60</td> <td>40</td> </tr> <tr> <td>S2</td> <td>5</td> <td>75</td> <td>65</td> </tr> <tr> <td>S3</td> <td>8</td> <td>90</td> <td>85</td> </tr> <tr> <td>S4</td> <td>1</td> <td>50</td> <td>30</td> </tr> <tr> <td>S5</td> <td>7</td> <td>85</td> <td>80</td> </tr> </table> The clustering is to be performed using the features: Study Hours and Attendance (%) You are given the initial centroids as: C1 = (2, 60) ,C2 = (7, 85) (a) Perform one complete iteration of the K-Means algorithm: (b) Based on the clusters formed, interpret each cluster in terms of student behavior (e.g., low engagement vs high engagement learners). (c) Compare the cluster grouping with the actual Marks: (d) Critically analyze the limitations of using unsupervised learning (clustering) in educational performance analysis. Consider aspects such as lack of labels, feature dependency, and interpretability. <table> <tr> <th>Student</th> <th>Distance to C1 (2,60)</th> <th>Distance to C2 (7,85)</th> <th>Assigned Cluster</th> </tr> <tr> <td>S1</td> <td>0.00</td> <td>25.49</td> <td>C1</td> </tr> <tr> <td>S2</td> <td>15.30</td> <td>10.20</td> <td>C2</td> </tr> <tr> <td>S3</td> <td>30.59</td> <td>5.10</td> <td>C2</td> </tr> <tr> <td>S4</td> <td>10.05</td> <td>35.51</td> <td>C1</td> </tr> <tr> <td>S5</td> <td>25.49</td> <td>0.00</td> <td>C2</td> </tr> </table> Final Clusters	Student	Study Hours (per day)	Attendance (%)	Marks	S1	2	60	40	S2	5	75	65	S3	8	90	85	S4	1	50	30	S5	7	85	80	Student	Distance to C1 (2,60)	Distance to C2 (7,85)	Assigned Cluster	S1	0.00	25.49	C1	S2	15.30	10.20	C2	S3	30.59	5.10	C2	S4	10.05	35.51	C1	S5	25.49	0.00	C2	10M			
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	Cluster Students C1 S1, S4 C2 S2, S3, S5 Updated Centroids Cluster New Centroid C1 (1.5, 55) C2 (6.67, 83.33)				
OR					
3b	Consider the following dataset with two features: X1 X2 2 4 3 6 4 8 5 10 Apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset from 2D to 1D. Project data onto PC1 Compute Mean: X1 : 3.5 Mean of X2: 7 Covariance Matrix=[1.67 3.33 3.33 6.67] Compute Eigenvalues and Eigenvectors $\lambda_1 \approx 8.34$ $\lambda_2 \approx 0$ Eigenvector for PC1: $v_1 = [0.447$ 0.894] Project Data onto PC1 Centered Point Projection on PC1 (-1.5, -3) -3.35 (-0.5, -1) -1.12 (0.5, 1) 1.12 (1.5, 3) 3.35 Final Reduced 1D Data [-3.35, -1.12, 1.12, 3.35]	10M			
OR					

4a	An agent is learning in an environment where the optimal actions may change over time. To balance exploration and exploitation, the agent adopts an ϵ-greedy action selection strategy. (a) Analyze how the ϵ-greedy strategy affects the agent’s ability to discover better actions over time, especially when the environment is changing. (b) Evaluate the impact of choosing a very small ϵ versus a very large ϵ on the quality of the learned policy. (c) Based on your analysis, discuss which choice of ϵ would be more suitable for such environments and justify your answer. Effect of ϵ-Greedy Strategy in a Changing Environment In the ϵ-greedy strategy: With probability $1-\epsilon$, the agent chooses the best-known action (exploitation). With probability ϵ, the agent chooses a random action (exploration). Analysis Exploration allows the agent to discover new or better actions. Exploitation helps the agent maximize reward using current knowledge. In changing environments, previously optimal actions may become suboptimal. Continuous exploration helps the agent adapt to environmental changes. Without exploration, the agent may remain stuck using outdated actions. Thus, ϵ-greedy improves adaptability while maintaining learning efficiency. Impact of Small ϵ vs Large ϵ <table><tr><th>Aspect</th><th>Very Small ϵ</th><th>Very Large ϵ</th></tr><tr><td>Exploration</td><td>Very low</td><td>Very high</td></tr><tr><td>Exploitation</td><td>Very high</td><td>Very low</td></tr><tr><td>Learning Speed</td><td>Faster initially</td><td>Slower</td></tr><tr><td>Stability</td><td>More stable</td><td>Less stable</td></tr><tr><td>Adaptation to Changes</td><td>Poor</td><td>Better</td></tr><tr><td>Risk</td><td>Gets stuck in local optimum</td><td>Excessive randomness</td></tr></table> Suitable Choice of ϵ for Dynamic Environments For environments where optimal actions change over time, a moderate or adaptive ϵ is most suitable. Recommended Approach Start with a relatively higher ϵ for exploration. Gradually reduce ϵ over time. Maintain a small nonzero ϵ to continue adapting to changes. This is often called a decaying ϵ-greedy strategy.	Aspect	Very Small ϵ	Very Large ϵ	Exploration	Very low	Very high	Exploitation	Very high	Very low	Learning Speed	Faster initially	Slower	Stability	More stable	Less stable	Adaptation to Changes	Poor	Better	Risk	Gets stuck in local optimum	Excessive randomness	10M			
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4b	The Q-learning update rule is: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$ Question: Analyze each component of the update rule and explain how it contributes to	10M																								

learning.

Further, discuss what happens in the following cases:

- $\alpha = 0$
- $\gamma = 0$
- very high α

Role of Each Component

1. Current Q-value $Q(s,a)$

- Stores existing knowledge.
- Acts as the baseline estimate before updating.

2. Immediate Reward r

- Encourages desirable actions.
- Positive rewards increase Q-values.
- Negative rewards reduce Q-values.

3. Discount Factor γ

- Controls importance of future rewards.
- Higher $\gamma \rightarrow$ long-term planning.
- Lower $\gamma \rightarrow$ focus on immediate rewards.

4. Maximum Future Q-value

$\max_a Q(s',a')$

- Chooses the best estimated future action.
- Helps the agent learn optimal policies.

5. Learning Rate α

- Determines update magnitude.
- Small $\alpha \rightarrow$ slow learning.
- Large $\alpha \rightarrow$ aggressive updates.

Case Analysis

Case 1: $\alpha=0$

Effect

$Q(s,a) \leftarrow Q(s,a)$

No update occurs.

Consequences

Agent does not learn from experience.

Q-values remain unchanged forever.

Learning process completely stops.

Case 2: $\gamma=0$

Update rule becomes:

$Q(s,a) \leftarrow Q(s,a) + \alpha[r - Q(s,a)]$

Effect

- Agent considers only immediate reward.
- Future rewards are ignored.

Consequences

- Short-sighted behavior.
- Suitable only when immediate rewards matter.
- Poor long-term decision making.

Case 3: Very High α

Effect

- New experiences dominate old knowledge.
- Q-values change rapidly.

Consequences

Advantages

- Faster adaptation.
- Quick learning in changing environments.

Disadvantages

- Learning may become unstable.

	• Q-values may oscillate heavily. • Sensitive to noisy rewards.				
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